

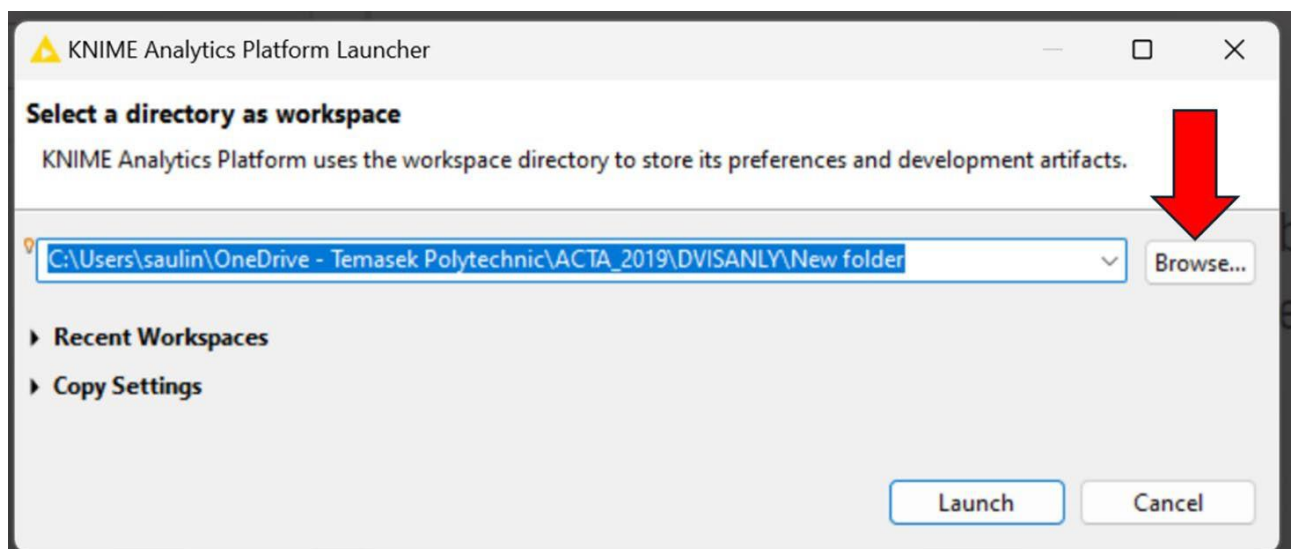
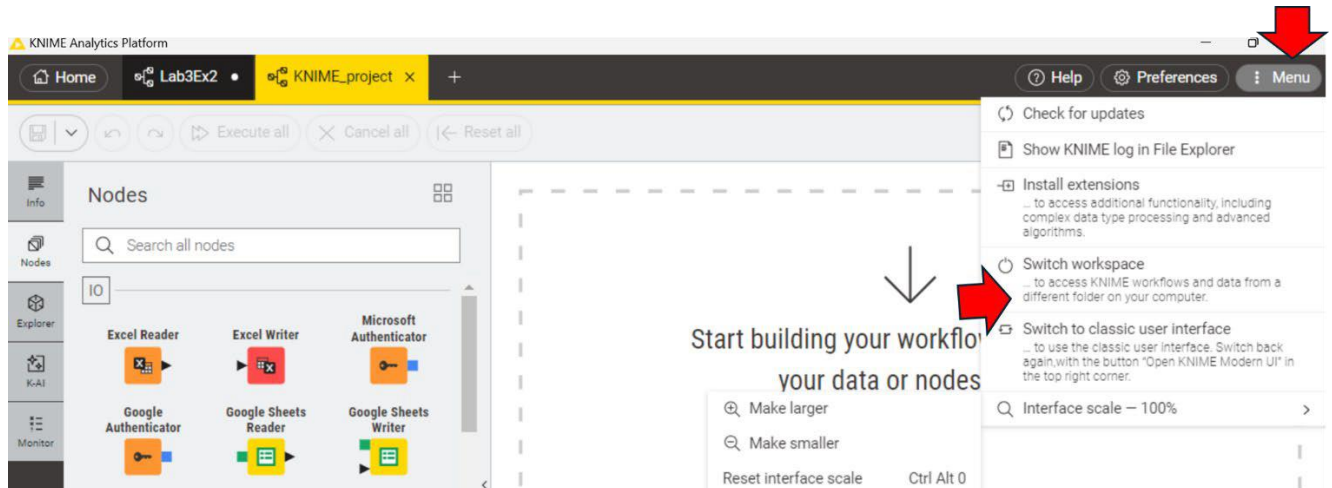
Lab 5 (Tutor Copy)

Learning outcomes:

1. Apply data cleaning and pre-processing for visualisation and analysis
2. Apply appropriate techniques to handle missing values
3. Present results of analytics visually using an appropriate platform
4. Interpret the data based on the visualisation to find possible relationships
between variables in the data

Tips on general workspace

Setting your workspace



The option in Menu, Switch Workspace, allow you to change your working directory (folder). Use "Browse" to change your directory.

Click on "**Launch**" to restart KNIME at selected directory.

Exercise 1

Data set: [tomslee_airbnb.csv](#)

Create a new workflow, named as “*Airbnb_data*”

This data is retrieved from : <http://tomslee.net/category/airbnb-data>

(You can refer to this site for the explanation of the variable used in the data)

Data dictionary (next page) explains the variables used in the data.

Attributes description of data file **tomslee_airbnb.csv**.

room_id:	A unique number identifying an Airbnb listing. The listing has a URL on the Airbnb web site of http://airbnb.com/rooms/room_id
host_id:	A unique number identifying an Airbnb host. The host's page has a URL on the Airbnb web site of http://airbnb.com/users/show/host_id
room_type:	One of "Entire home/apt", "Private room", or "Shared room"
borough:	A sub region of the city or search area for which the survey is carried out. The borough is taken from a shapefile of the city that is obtained independently of the Airbnb web site. For some cities, there is no borough information; for others the borough may be a number. If you have better shapefiles for a city of interest, please send them to me.
neighbourhood:	As with borough: a sub region of the city or search area for which the survey is carried out. For cities that have both, a neighbourhood is smaller than a borough. For some cities there is no neighbourhood information.
reviews:	The number of reviews that a listing has received. Airbnb has said that 70% of visits end up with a review, so the number of visits. Note that such an estimate will not be reliable for an individual listing (especially as reviews occasionally vanish from the site), but over a the number of reviews can be used to estimate city as a whole it should be a useful metric of traffic
overall_satisfaction:	The average rating (out of five) that the listing has received from those visitors who left a review.
accommodates:	The number of guests a listing can accommodate.
bedrooms:	The number of bedrooms a listing offers.
price:	The price (in \$US) for a night stay. In early surveys, there may be some values that were recorded by month.
minstay:	The minimum stay for a visit, as posted by the host.
latitude and longitude:	The latitude and longitude of the listing as posted on the Airbnb site: this may be off by a few hundred metres. I do not have a way to track individual listing locations with
last_modified:	the date and time that the values were read from the Airbnb web site

Create a new workflow, named “tomslee_missingValue”.

Use **CSV Reader** node to read [tomslee_airbnb.csv](#)

CSV Reader

Add comment

► 1: File Table

Flow Variables

Rows: 2815 | Columns: 14

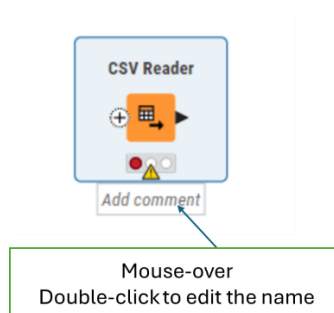
TableStatistics

#	Row...	room_id Number (int...	host_id Number (int...	room_ty... String	borough String	neighbo... String	reviews String	overall_... Number (do...	a Ni
1	Row0	30423	129623	Private room		TS20	0		2
2	Row1	50620	231938	Entire home/...		TS28	0		
3	Row2	56334	266763	Private room		MK13	19	5	1

How many observations are there? **2815**

How many variables? **14**

Tips: How to name your node?



Select the node, mouse over the “Add comment” and double-click on it.

Step 1: Any error in data type assignment?

Before handling the missing values, check if there is any inaccurate assignment of data type?

► 1: File Table

Flow Variables

Rows: 2815 | Columns: 14

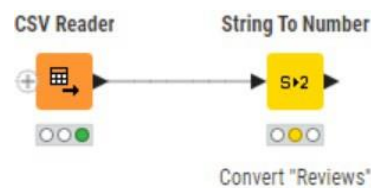
Table

Statistics

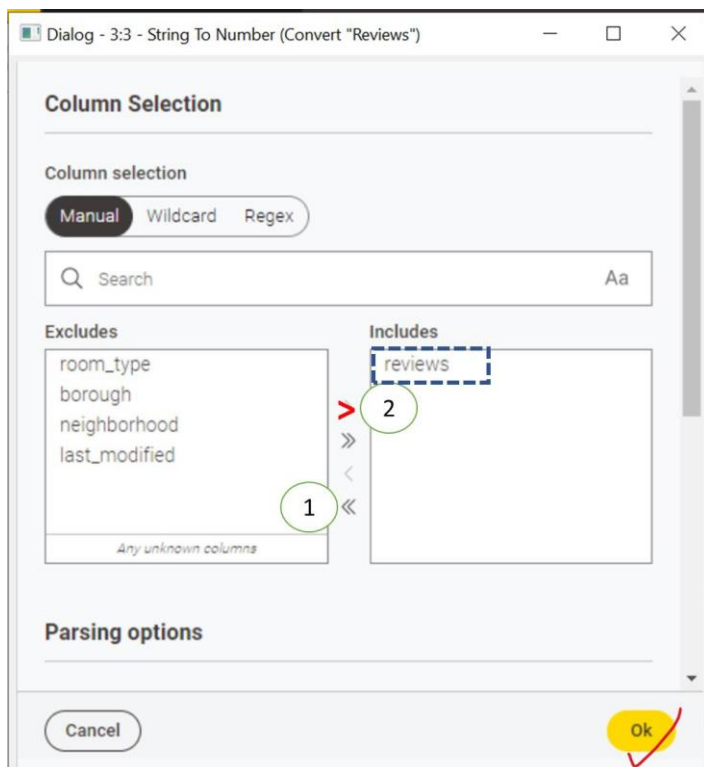
#	Row...	room_id Number (int...	host_id Number (int...	room_ty... String	borough String	neighbo... String	reviews String	overall_... Number (do...	a Ni
1	Row0	30423	129623	Private room	TS20		0		2
2	Row1	50620	231938	Entire home/...	TS28		0		
3	Row2	56334	266763	Private room	MK13		19	5	1
4	Row3	69694	350259	Private room	TS21		0		1
5	Row4	252903	1328128	Entire home/...	MK25		10	5	2
6	Row5	71609	367042	Private room	MK28		8	4.5	

Shouldn't this be Number? Why String?

Add a **"String To Number"** node to convert the data type to Number.



Configure **"String To Number"** node:



Execute the node. Check the output table from **“String To Number”**:

Error Message

"e" in cell [Row40, column "reviews", row 41] can not be transformed into a number

RowID	neighborhood	reviews	overall_sat...
Row38	MK28	7	4.5
Row39	MK17	8	?
Row40	MK83	e	4.5

For input string: "e"

So, there is a character “e” at row 40, causing the variable to be interpreted as “String”.

This character “e” is replaced as missing value.

Successful conversion. The character is now treated as “missing value”

Rows: 2815 | Columns: 14

#	Row...	room_id	host_id	room_ty...	borough	neighbo...	reviews	overall_...	a
1	Row0	30423	129623	Private room	TS20	0	?	2	
2	Row1	50620	231938	Entire home/...	TS28	0	?	?	
3	Row2	56334	266763	Private room	MK13	19	5	1	
4	Row3	69694	350259	Private room	TS21	0	?	1	
5	Row4	252903	1328128	Entire home/...	MK25	10	5	2	
6	Row5	71609	367042	Private room	MK28	8	4.5	?	
7	Row6	71896	367042	Private room	MK28	15	4	3	
8	Row7	71903	367042	Private room	MK28	13	4.5	3	



What about the date&time in the last column?

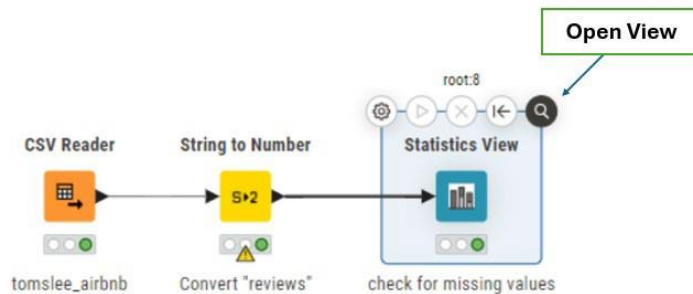
The last column “Last-modified” should be Date&Time format, but it was interpreted as “String”.

As we are doing analysis in TABLEAU, and TABLEAU interpret the date&time correctly. So, we will not do String to Datetime transformation.

The method of String to DateTime will not be tested.

Step 2: Identify where is the missing value

Add **Statistics** node to the file Table output.



No configuration needed. Run the node (turn to green traffic light). Check out the result (Open View) or table view (below the selected node).

Interactive View: Statistics View

Statistics

Rows: 14 | Columns: 14

Mouse-over
Sort descending

Name	Type	# Missing va...	↓	# Unique val...
borough	String	2815		0
overall_satisfac	Number (double	1110		8
accommodates	Number (double	361		9
minstay	Number (double	73		16
bedrooms	Number (double	4		7
reviews	Number (double	1		73
room_id	Number (intege	0		2815
host_id	Number (intege	0		1424

? How many missing values?

Statistics node should be able to show how many missing values.

Name of the variable	Number of missing values
reviews	1
overall_satisfaction	1110
accommodate	361
bedrooms	4
minstay	73
borough	2815

List of missing values

We need to know in total how many missing values, to decide our strategy to discard the columns or rows.

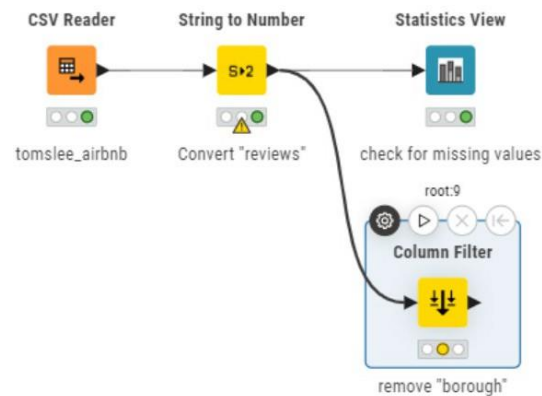


Discard row reduces the number of observations. Discard column loses a variable for analysis.

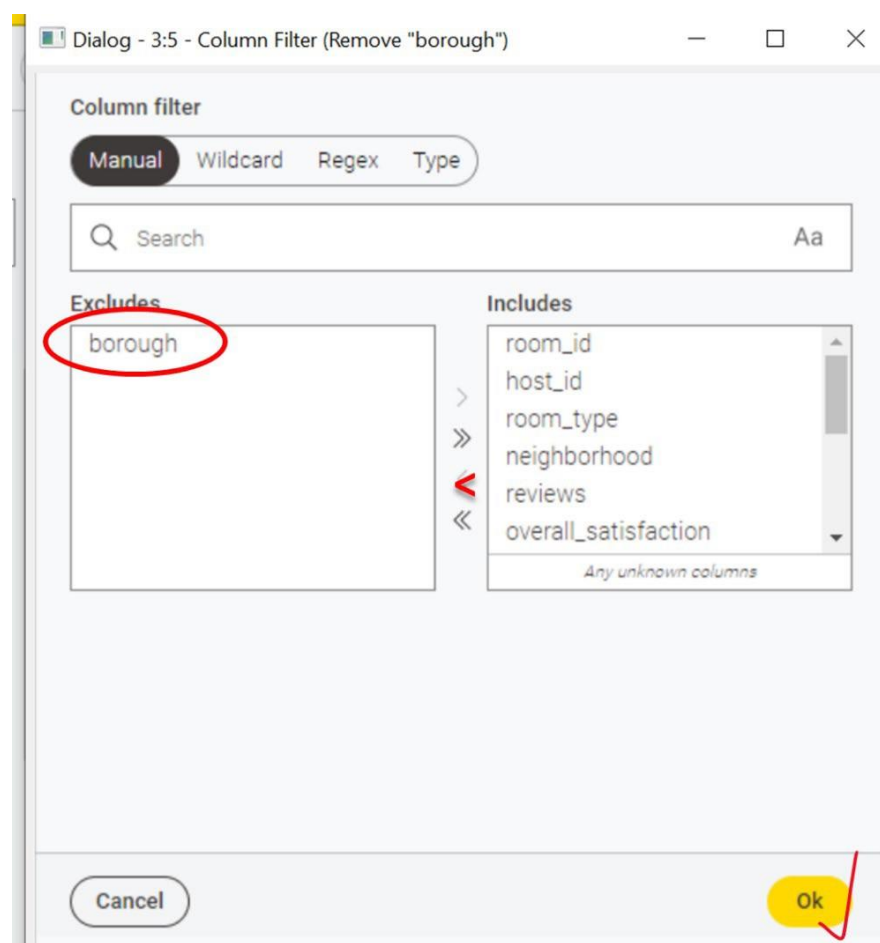
Notice the variable “*borough*” is useless because all values are missing!

In such case, need to remove the column “borough”.

Add “**Column Filter**” node after the output table:

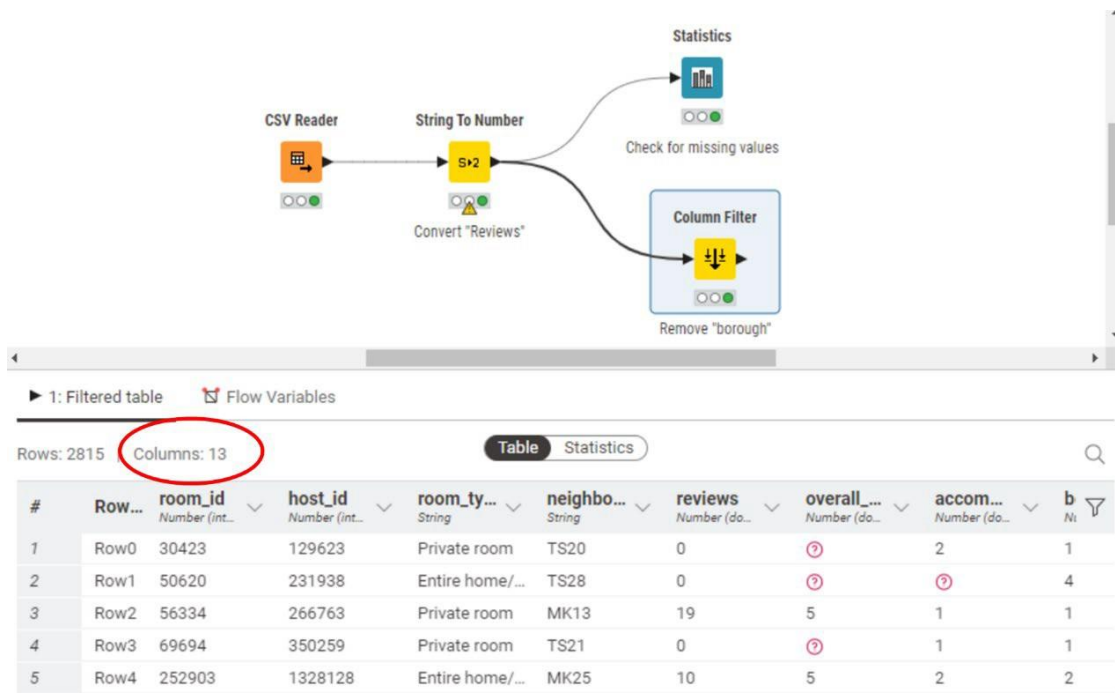


Configure the **Column Filter**:



Only exclude “borough” column.

Run (execute) the Column Filter node.



One column, “borough” has been removed. Number of columns is now 13.



Alternate method:

There are 2 possible nodes to use:

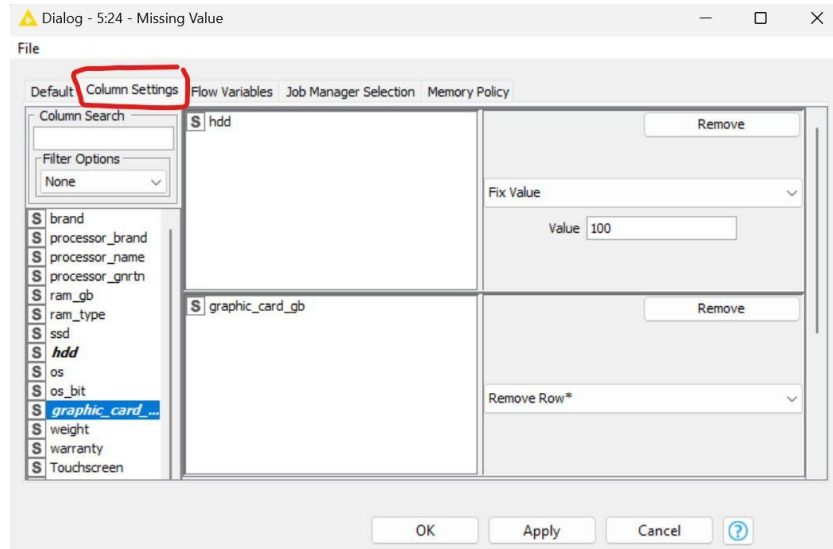
Missing Value Column Filter, or Missing Value



[Missing Value Column Filter] node removes all **columns** from the input table which contain more missing values than a certain percentage.

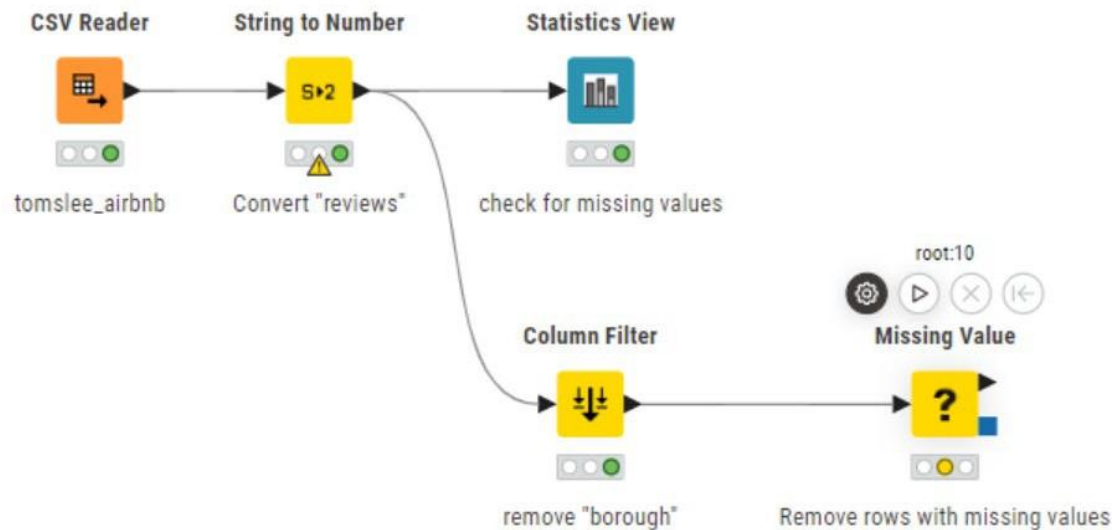


[Missing Value] node deal with **rows** operation. It has option to remove rows with missing values or replace missing values with other values. This method is more direct, when you know which columns to remove.

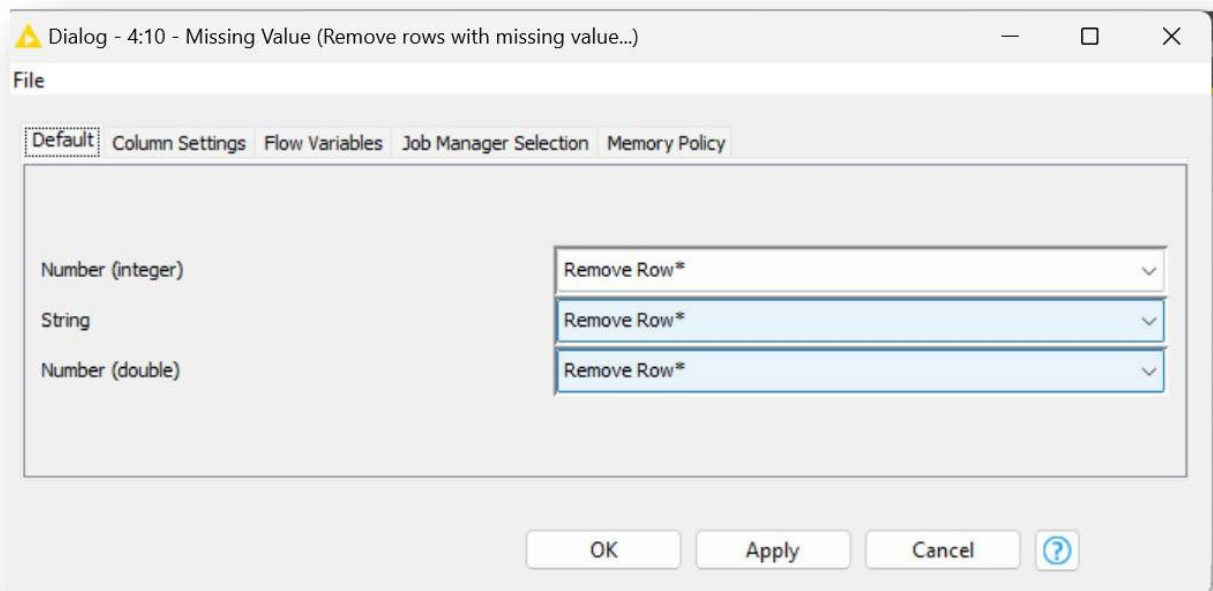


How to handle the rest of the missing values?

Simple way is to **remove rows with missing value**. This is easily done by “**Missing Value**” node.



Make sure you configure the **Missing Value** node:





Notice the asterisk (*) for “**Remove Row**” option. This means this option will not produce valid PMML 4.2.

*PPML – Predictive Model Markup Language

This is a file exchange format. It means the file becomes non-standard PMML.

As we are not going to export to PMML format, *we can ignore.*

To know more about the various options, click on the question mark (?) for information.

Run (execute) the “Missing Value” node.

Check out the final table:

#	RowID	room_id	host_id	room_type	neighbor...	reviews	ove
1	Row2	56334	266763	Private room	MK13	19	5
2	Row4	252903	1328128	Entire home/ap	MK25	10	5
3	Row6	71896	367042	Private room	MK28	15	4
4	Row7	71903	367042	Private room	MK28	13	4.5
5	Row9	71915	367042	Private room	MK28	11	4

Total observations are 1444.

Number of observations lost = 2815 – 1444 = 1371

The sample size is large enough to use for analysis.

For a quick view of the statistics result (without adding another statistics node):

Name	Type	# Missing values	# Unique valu...	Minimum
room_id	Number (intege	0	1444	56,334
host_id	Number (intege	0	741	23,666
room_type	String	0	3	?
neighborhood	String	0	58	?

The “Missing Values” column shows all “0”. There are no more missing values in the data table.



What happen if we reverse the order:

use **Missing Value** node before **Column Filter**?

If we remove rows containing missing values before removing “borough” column, we will be left with no rows. No data.

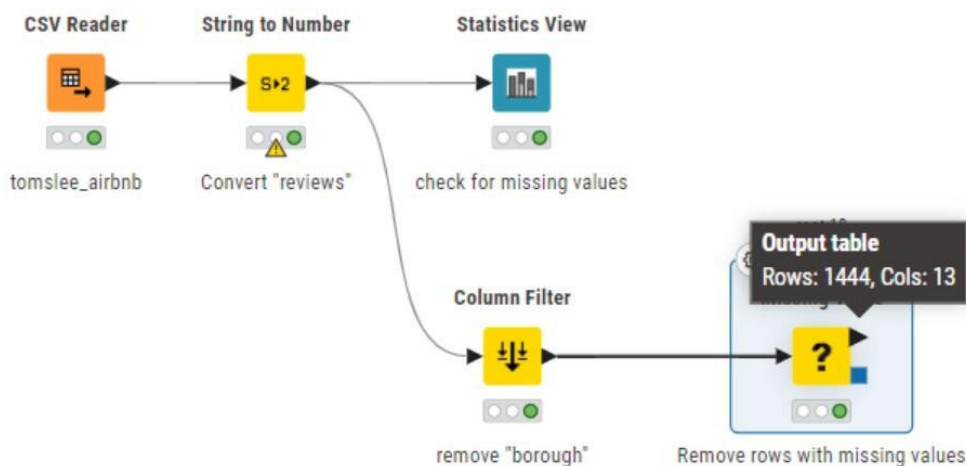
All the rows will be removed because all the rows contain missing value in “borough” variable.

So, we should remove the column “borough” first.

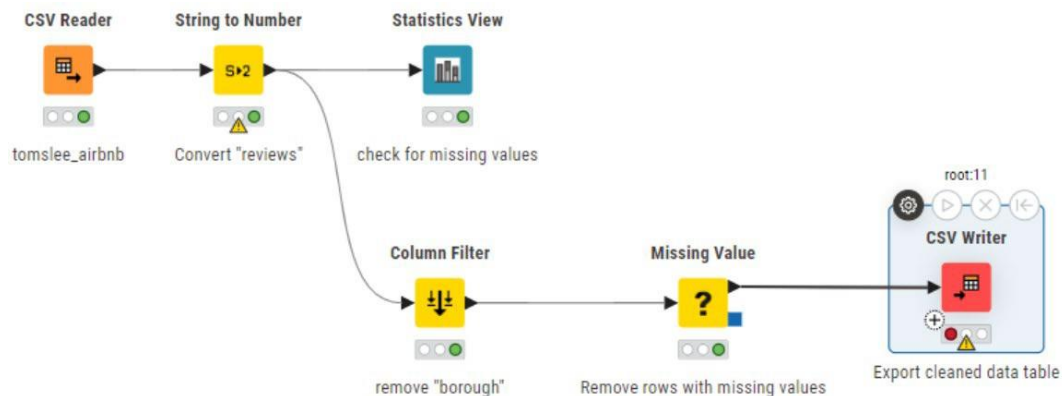
How to transfer this cleaned data set to TABLEAU?

The data is cleaned now. We will export this cleaned data set to use in TABLEAU.

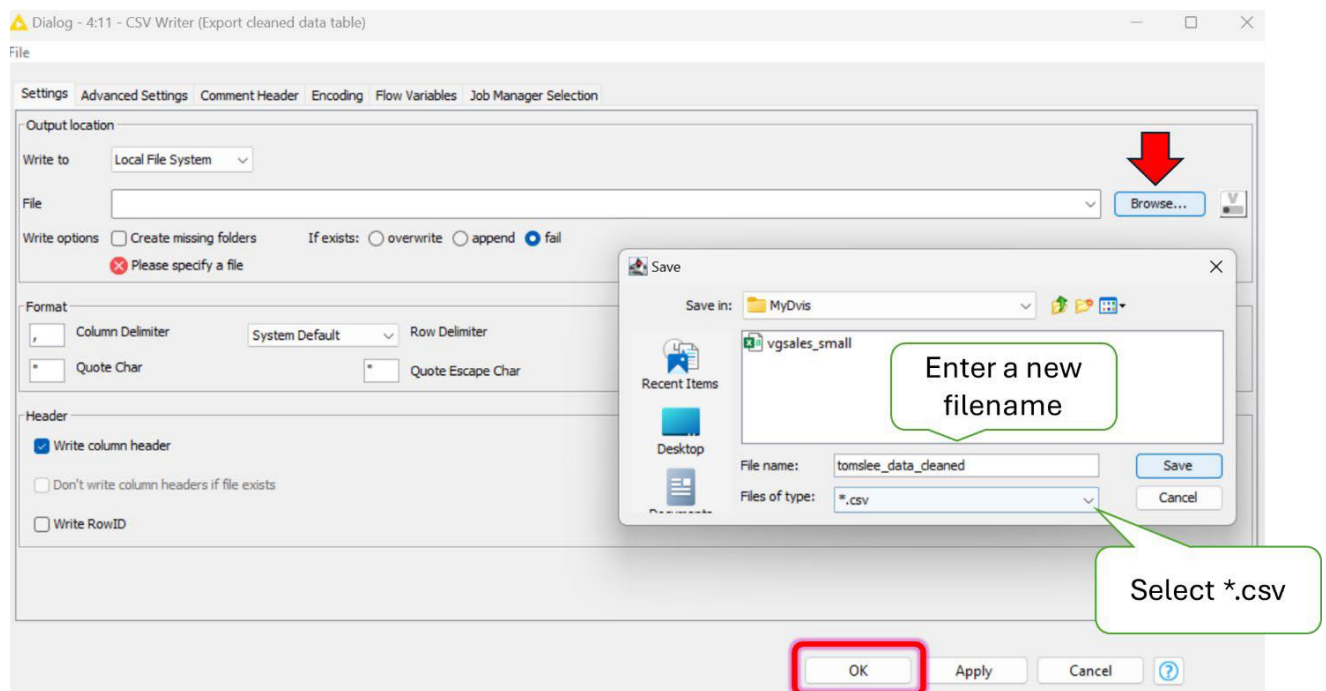
The cleaned data set is in the output node of Missing Value node.



Add “**CSV Writer**” to a save the final data table to a new file for analysis in TABLEAU.



Configure the **CSV Writer** node:



This time, “browse” to your preferred folder (“MyDvis”) to save the cleaned data. You will need to give a filename for this cleaned data. Remember the filename you have given, as you will read this file into TABLEAU.

Remember to run (execute) the CSV Writer node, so that a new file will be generated.

Use file explorer to check that the new file is created in your selected folder. This file will be used in Exercise 3.

Saved your KNIME file.



Exercise 2: On your own

Data set: *personal_info_with_missing_data.csv*

There are 12 rows and 9 columns.

There are missing values, in-consistence values (n/a, NA, empty) that means “no”; as well no header name for “Col0”.

How can you “clean” the data so that they useful for analysis?

For example:

Replace missing value in “**age**” with Mean.

Replace missing value in “**year seniority**” with “Most frequent value”.

The “**parking space**” is a running number. Look like number 12 is missing.

Replace the only missing value in “**parking space**” with “12”.

Replace missing value in “**emergency_contact**” with “no”.

Replace missing value in “**attending_party**” with “0”.

Remove the variable “**pets**” as there are too many missing values.

(You may refer KNIME workbook from LMS for extra nodes information)

Show your workflow to your tutor.

Replace missing value in “age” with Mean.

Replace missing value in “year seniority” with “Most frequent value”.

The “parking space” is a running number. Look like number 12 is missing. Replace the only missing value in “parking space” with “12”.

Replace missing value in “emergency_contact” with “no”.

Replace missing value in “attending_party” with “0”.

Remove the variable “pets” as there are too many missing values.

[\[Show Me\]](#)

Exercise 3 : More practice on EDA (Submit to LMS)

Use the clean data in Exercise 1 ([tomslee_data_cleaned.csv](#)) to perform Exploratory Data Analysis.

Read this clean data into TABLEAU.

Answer the following questions:

1. How many surveyance gave “Overall Satisfaction” ≥ 4 ?
(*plot bar chart of Count Vs “overall satisfaction” in TABLEAU)

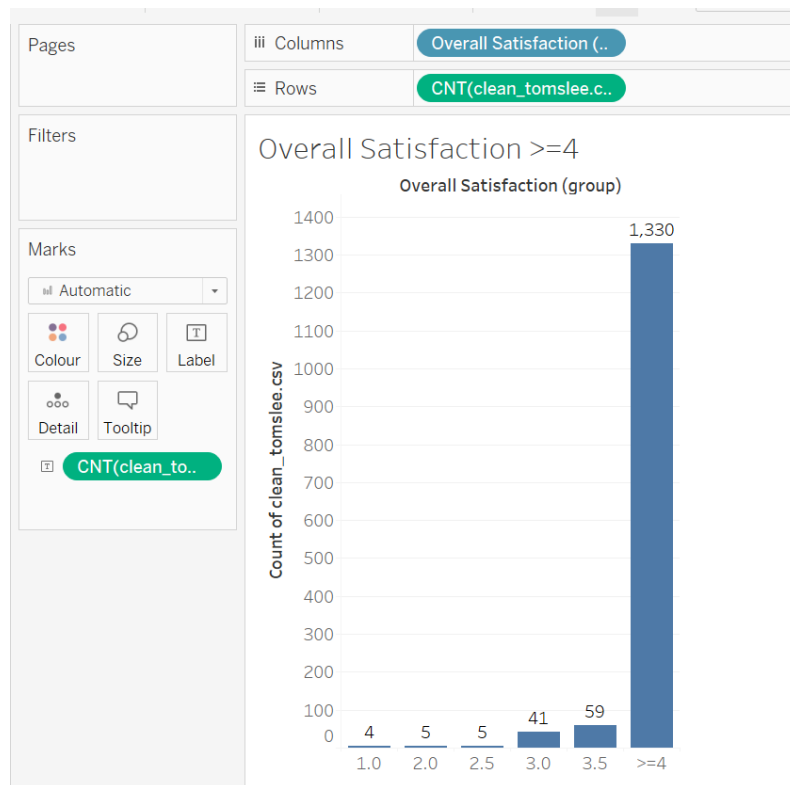
Firstly, the Overall Satisfaction variable need to be a Dimension.

Drag the variable up to the “Dimension” section.

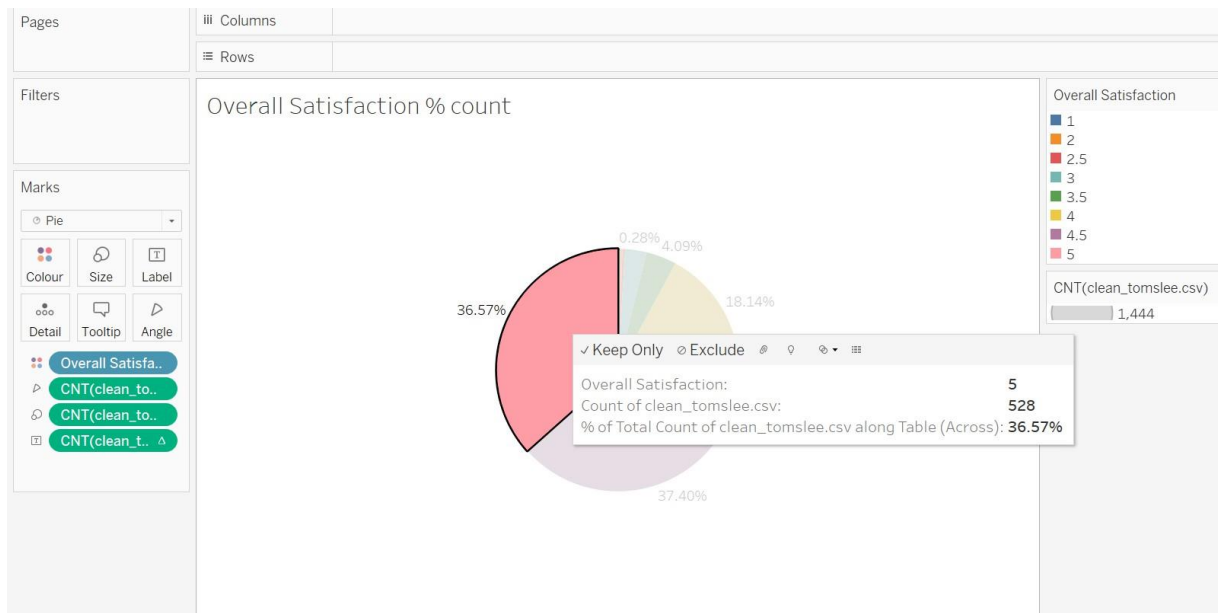
Tables

- # Host Id
- Last Modified
- Abc Neighborhood
- # Overall Satisfaction
- Overall Satisfaction (group)
- # Room Id
- Abc Room Type
- Abc Measure Names
- # Accommodates
- # Bedrooms
- Latitude
- Longitude
- # Minstay
- # Price
- # Reviews
- # clean_tomslee.csv (Count)
- Measure Values

[\[Show me\]](#)



2. What is the percentage of responses with “Overall Satisfaction” = 5?
(*plot a pie-chart)



[Show Me]

3. What is the distribution of the “price”?

(hint: support with box-plot and histogram)



There are outliers in the price.

The histogram distribution is skewed to the right.

[\[Show Me\]](#)

4. Data Driven Decision:

How much do I need to spend to find a comfortable room for one night?



Mr Lim (Finding a room)

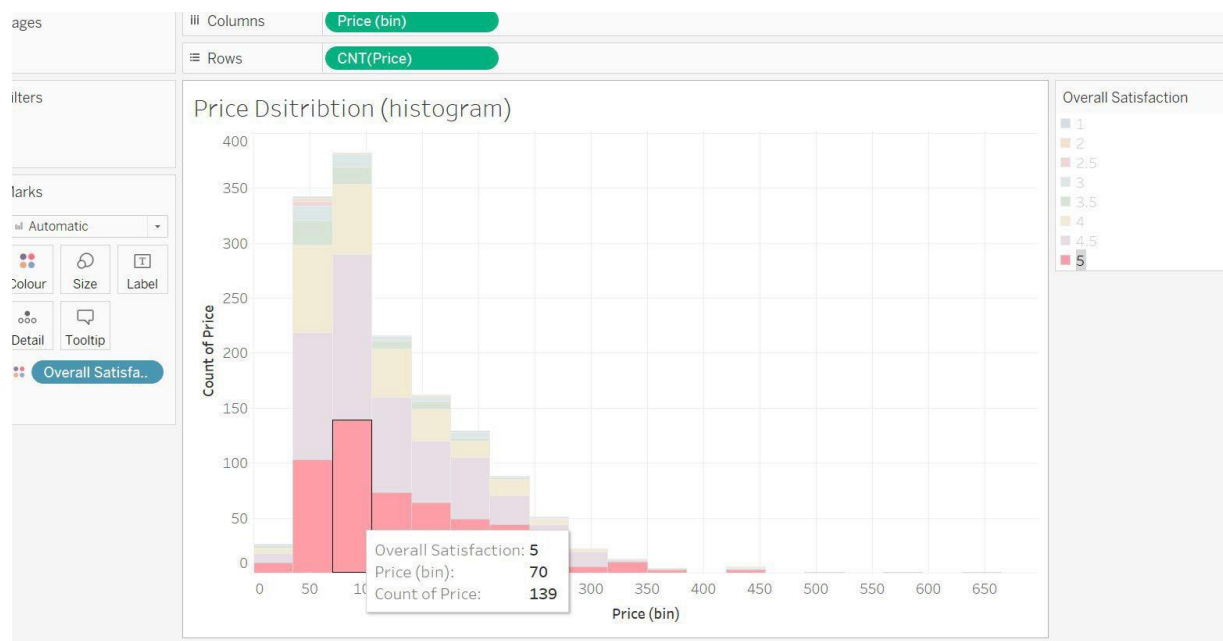
How would you advice?

There are many possible ways you can advise Mr Lim.

You can check his budget. And find what is the minimum and maximum price from your data sheet.

You can filter by type of room, or even location.

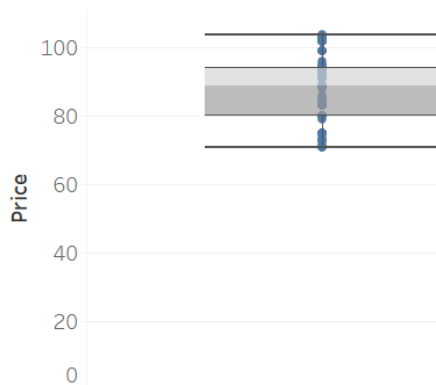
Below is based on Satisfaction level and most visited rooms.



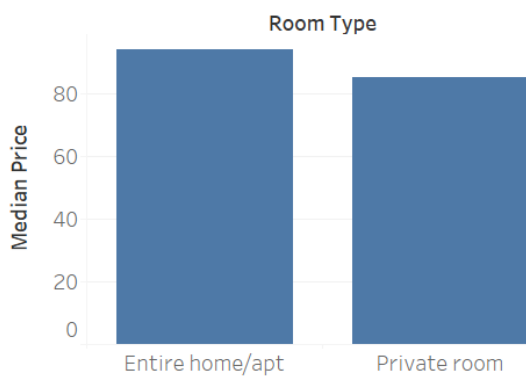
Based on overall satisfaction of 5, most units are priced between 70 to 100.

With this range of price, Mr Lim can get a private room at lower budget.

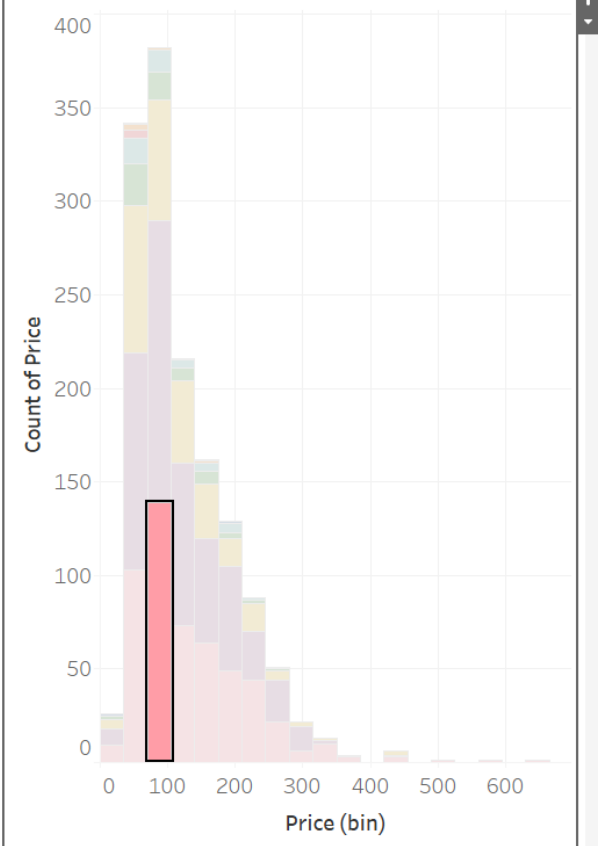
Price box plot



Median Price by Room Type



Price Distribution (histogram)



Project: Data Cleaning

You can start working on the data given to you.

Check if there is any missing values. Post screen-shot of relevant evidence in your report.

Explain how you will handle these missing values (Replace with ... or remove rows with missing values, or column filter or row filter, etc)

Check if there is any need for data transformation due to wrong data type (eg. Row Filter or String to Number, or Number to String or Category to Number, etc)

Show screen shot of the workflow.

Explain the impact (consequences) of the action taken to clean your data.

State whether there is any change in the number of observations and variables after cleaning.

Show your work in the next lesson.